INCH-POUND

MIL-P-63313 (AR)
AMENDMENT 7
19 August 1993
SUPERSEDING
AMENDMENT 6
16 MARCH 1993

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M80A1
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63313 (AR), dated 29 June 1979, and is approved for use by the U. S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1: Delete "(Alternate)".

2.1:

* Add:

"MIL-A-70625 - Automated Acceptance Inspection Equipment Design, Testing and Approval, of."

MIL-E-46215: Delete "Igniter, Electronic Primer, M83 and Electric Primers, M80A1 and M83" and substitute "Electric Primers, M80A1, M83 and M120".

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PAGE 2

2.1:

Add: "MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes".

MIL-STD-286: Delete "Propellants: Sampling, Inspection and Testing" and substitute "MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing".

Drawings: 8839499: Delete "Sheet 2-".

3.1.1: Delete "(Sheet 2)".

3.3.1: Delete "...at the loading station at the time of loading the ignition element...", and substitute "...after drying...".

Delete "...0.10 percent." and substitute "...0.02 percent."

PAGE 3

* 3.8.1: Delete:

"3.8.1 Assembly torque. Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet."

and substitute:

"3.8.1 Head loading assembly. The Retainer and Closing Plug Loading Assembly shall comply with drawing torque requirements. (See 4.5.8.1)".

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* 3.8.2: Delete:

"3.8.2 Disassembly torque. Threaded joints shall withstand application of the minimum disassembly torque specified on the applicable drawing, without moving, after thread sealing compounds has cured."

and substitute:

"3.8.2 Primer loading assembly. The Primer Head/Body threaded joint shall comply with drawing torque requirements. (See 4.5.8.2)".

PAGE 4

4.2: Delete "at" and substitute "on".

4.3.1: Delete "(Sheet 2)".

PAGE 6

* TABLE I:

For Head loading assembly: Delete "4.4.2.3 &".

Examination or Test Block: Delete "-Sheet 2".

PAGE 8

4.4.2.1: Note 1: Delete in its entirety and substitute:

"Three samples minimum shall be selected from each drying oven after completion of drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved or rejected. The selected samples must not be dried more than any other unit which could be selected."

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* 4.4.2.3: Delete in its entirety.

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* 4.4.2.4:

Delete in its entirety and substitute new paragraph 4.4.2.4 included herein as page 5.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Head Loading Assembly	1 OF 1	9327345
			NEXT HIGHER ASSEMBLY: 8839499

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-functioning test		4.4.3.1.2	3.5.2	4.5.5
Major					
101	Closing disk missing or damaged exposing explosive		0.40%	3.2	Visual or gage
102	Retainer torque failure		0.40%	3.8.1	4.5.8.1
103	Retainer lacquer or stake missing	1/	0.40%	3.2	Visual
104	Functioning test		4.4.3.1.1	3.5.1	4.5.4
105	Electrical resistance test		0.40%	3.4	4.5.3
106	Depth to electrode		0.40%	3.2	Gage
107	Flash hole obstructed		0.40%	3.2	Gage
108	Closing plug torque failure		0.40%	3.8.1	4.5.8.1
109	Lacquer missing from closing plug threads	1/	0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: Inspections requiring access to inner components and inspections conducted by automatic assembly machine may be performed before completion of entire Head Loading Assembly.

1/ For automatic machine assembly, disassemble 1 unit/hour and visually inspect. Absence of lacquer/stake or inadequate lacquer application will cause rejection of all production since last acceptable sample was inspected.

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4.4.2.5: Drawing Number Block: Delete "(sheet 2)".

Add Critical Defect:

"2 Plug missing 100% 3.2 Visual".

Category 201: Delete "Visual" (appearing in Amendment 5) and substitute "Gage" (as in the Basic Specification).

PAGE 19

4.4.3.2.1: Delete "Beginning with the first lot produced and continuing until three " and substitute "Beginning with the first lot produced, or the lot following a rejected lot in 4.4.3.2.2, and continuing until three".

4.4.3.2.2: Add to the end of the paragraph: "If a lot is rejected under these criteria, continue testing lots in accordance with 4.4.3.2.1."

4.5.2.1: Delete in its entirety and substitute:

"4.5.2.1 Igniter charge composition. The moisture content of the igniter charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour drying time shall be used. The upper insulator from the electric ignition element loading assembly may be removed prior to analysis."

4.5.2.1.1: Delete in its entirety.

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4.5.2.3: Delete "Method 102.1" and substitute "Method 102.1.3".

* 4.5.3:

Line four: Delete "or" and substitute "of".

Delete "- sheet 2".

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4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3, and 4.5.6.4: Delete in their entirety and substitute the following:

"4.5.6 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate & Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2%
Charcoal	20.0 ± 1.5%

4.5.6.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70° C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5° C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)\ (100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.5.6.2 Lead Thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

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Percent lead thiocyanate = $\frac{16.17\ (AB-CD)}{E}$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.5.6.3 Potassium chlorate and gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

PAGE 22

* 4.5.8.1: Delete in its entirety and substitute:

"4.5.8.1 Head loading assembly. Torque shall be verified by either performing torque inspections of assembled components or by periodic verification of the torque machine. When torque is verified by inspection of assembled components, torque tests shall be performed by reapplication of torque, while the lacquer is still wet (within 5 minutes of application), and with the part suitably marked and held for detection of movement between adjacent parts. Any discernible movement between adjacent parts is a torque test failure. When torque application is confirmed by monitoring the torque machine, a periodic verification of machine output values is required. Detailed requirements of the verification technique shall be in accordance with MIL-A-70625A and 4.4.4. Failure to meet the requirement will cause rejection of all production since the last successful verification."

5.1, 5.1.1, 5.2 and 5.2.1: Delete in their entirety and substitute:

"5.1 Packaging and marking. Primers shall be packaged and marked in accordance with drawing 8837899.

5.2 Packing and marking.

5.2.1 Level A. Packaged primers shall be packed and marked in accordance with drawing 8837900.

5.2.2 Level B. Packaged primers shall be packed and marked in accordance with drawing 8837900."

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6.1: Delete "M392" and substitute "M392A2".

Delete "M724" and substitute "M724A1".

6.5.1: Delete "Method 102.1" and substitute "Method 102.1.3".

6.5.2: Delete in its entirety and substitute:

"6.5.2 Electrical resistance. This test shall be performed in accordance with the applicable defect table. Any primer which fails to comply with the requirements specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3."

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Figure 1: Title: Delete "PAINT" and substitute "LACQUER".

6.7: Delete "M392" and substitute "M392A2".

Delete "M724" and substitute "M724A1".

6.8: Delete "DRDAR-QAR-E and DRDAR-LCU-CV" and substitute "SMCAR-QAT-A and SMCAR-LCE-D".

Delete "01201" and substitute "61299".

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